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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCIAA201 - MATERIAL SPECIFICATIONS - I
(2024 ADMISSION ONWARDS)

Part A

1. 6.6 packets of cement

2. Sound Knot
   Unsound or Loose Knot
   Encased Knot
   Dead Knot

3. * It is very cheap and budget and friendly
   * ECO friendly product
   * Easy to clean and Easy to apply

4. A higher silica content in glass results in a higher hardness. Hardness is primarily determined by the strength and density of the glass's molecular network

5. An English bond is a bricklaying pattern that consists of alternating courses of headers and stretchers. A course is a single row of bricks. This arrangement creates one of the strongest and most commonly used bonds for constructing load-bearing brick walls.

6. It is a type of stone popular and cost-effective building material known for its durability, natural aesthetic, and versatile applications, especially for flooring

7. Terrazzo flooring is a composite material created by embedding chips of materials like marble, quartz, granite, or glass into a binder. This mixture is then poured, cured, and ground down to create a smooth, polished, and decorative surface.

8. Beeswax Polish – Matt / Satin finish
   French Polish – High glossy finish

9. Rock wool (or mineral wool) plays a crucial role in acoustic control by acting as both a sound absorber and a sound barrier. Its effectiveness stems from its dense, fibrous, and porous structure, which disrupts and dissipates sound energy.

10. Two key defects that significantly reduce the durability of paint are **peeling/flaking** and **chalking**. These defects compromise the protective qualities of the paint film and can lead to damage of the underlying surface

11. Lime cement is a mixture of lime and cement used as a binding material in construction. It combines the workability and smooth finish of lime with the strength and quick setting of cement. This mix is often used for plastering and masonry work to improve durability and bonding.

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12 Plain or ordinary concrete, Reinforced concrete (RCC), M20 grade concrete?

Part B

13 **Defects of paint**

Peeling and Blistering
Blistering (oil/grease) or **peeling** (moisture) results from air bubbles under the paint. Causes include excessive layers, poor surface preparation, high humidity, or incompatible paint
Prevention & Repair: Use porous paints like emulsions, scrape defective areas, apply a primer, and repaint

Chalking is the formation of powdery residue mainly caused by insufficient oil, inferior paints, or sun exposure
Remedies: Use adequate primers, lighter shades, and water-based paints; remove chalking by washing and re painting if necessary.

Flaking occurs when the paint film detaches due to poor adhesion or moisture ingress
Remedies: Proper surface cleaning and sanding before paint; use compatible primers, then repaint loose areas
Cracking/Checking: Small fractures due to environmental stress or early recoating .

14. **Difference between english bond and flemish bond**
English bond is stronger with simple alternating courses of stretchers and headers, while Flemish bond alterna tes headers and stretchers in each course, offering better visual appeal but slightly less strength.
Bond Pattern: English bond has alternating courses of **stretchers** and **headers**, whereas Flemish bond has **al l courses consisting of alternating stretchers and headers**
Strength: English bond is **stronger** and ideal for load-bearing walls thicker than 1.5 bricks; Flemish bond is co mparatively **weaker** and better suited for walls where aesthetics matter
Appearance: English bond has a **plain and solid look**, while Flemish bond provides a **decorative, symmetric al, and attractive appearance**
Skill Required: English bond is **easier to construct** and requires less supervision, while Flemish bond deman ds **skilled labor** due to precise alignment of bricks and breaks in vertical joints
Applications and Cost: English bond is used for **foundations, structural, and thick load-bearing walls** and i s often more **expensive**; Flemish bond is preferred for **exposed walls, façades, and decorative walls**, and is relatively **economical**.

15. Plaster, primer, putty, first coat of paint, second coat of paint

17. In woodworking, a **lengthening joint** is a type of joint used to connect two pieces of timber lengthwise to create a single, longer member. These are also known as **longitudinal joints** or **spliced joints**. They are a fundamental technique in carpentry and construction, used for everything from simple frames and scaffolding to structural beams in timber-framed buildings

Explanation or diagram of

Fished joint 2

Fished joint 2

Fished joint 3

Scarfed joint

Tables joint 1

Tables joint 2

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Lapped joint 1

Lapped joint 2

18. **Oil varnish**

Oil varnishes are a combination of natural or synthetic resins dissolved in a drying oil, like linseed oil.

Characteristics: Dries slowly but forms a very hard, elastic, and durable film. Oil varnishes are also highly resistant to water and weathering.

Uses in building works:

External woodwork: Ideal for exterior wood surfaces like doors, window frames, and decorative features that require maximum durability and weather resistance.

Structural members: Used on structural timber components that need a hard, long-lasting protective coat.

High-traffic floors: Premium floor varnishes often use an oil base for their high abrasion resistance.

Spirit varnish

Spirit varnishes are made by dissolving natural resins, such as shellac or lac, in a volatile solvent like methylated spirit.

Characteristics: Dries very quickly, provides a bright, glossy finish, but is not as durable or weather-resistant as oil varnish.

Uses in building works:

Interior woodwork: Primarily used for finishing high-quality interior woodwork and furniture where a quick-drying, aesthetic finish is desired.

French polishing: A classic finishing technique that uses shellac-based spirit varnish to create a high-gloss, smooth surface.

Sealer: Sometimes used as a clear sealant over knotty wood to prevent resin from bleeding through later paint layers.

Water-based (acrylic) varnish

Also known as acrylic varnish, this is a modern, eco-friendly option that uses synthetic resins with water as the solvent.

Characteristics: Fast-drying, low-odor, non-yellowing, and easier to clean up than solvent-based varnishes. It is also less toxic and more environmentally friendly.

Uses in building works:

Interior finishes: Ideal for interior applications, including wooden floors, doors, and trim, where low volatile organic compounds (VOCs) are a priority.

Woodwork and paneling: Provides a durable and clear finish for wooden furniture, cabinets, and paneling, enhancing the wood's natural grain.

Decorative surfaces: Used on a variety of decorative surfaces, including painted surfaces and some types of plaster, to provide a protective, transparent topcoat.

Polyurethane varnish

This is a synthetic varnish made with polyurethane resins, often combined with a traditional oil or water-based formula.

Characteristics: Extremely durable, highly resistant to abrasion, chemicals, and water, and available in both single- and two-component systems. Can be formulated for either oil-based (more durable) or water-based

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(fast-drying) use.

Uses in building works:

High-traffic areas: The superior toughness makes it the standard choice for wooden flooring in residential and commercial buildings.

Countertops and furniture: Used on surfaces that experience heavy use and require maximum resistance to scratches and spills.

External doors and windows: Provides exceptional weatherproofing and durability for exterior wood surfaces.

19. To reduce solar heat gain in a commercial building, I would use tinted and reflective glass in the glazing design. **Tinted glass** contains small amounts of color like grey, bronze, or blue, which helps absorb and reduce sunlight and heat entering the building. It also lowers glare and makes the indoor lighting softer and more comfortable. **Reflective glass**, on the other hand, has a thin metallic coating that reflects a large portion of the solar radiation away from the surface. This prevents heat from entering and keeps the interiors cooler, especially in hot climates.

20. Wooden Surfaces:

1. Preparation:
   - Sand the surface to smooth out any rough spots.
   - Fill any holes or cracks with wood filler.
   - Clean the surface to remove dust, dirt, or grease.
2. Priming (optional but recommended):
   - Apply a wood primer suitable for your paint type.
3. Painting:
   - Apply paint suitable for wood, using a brush or roller.
   - Follow the manufacturer's instructions for application and drying times.

Metal Surfaces:

1. Preparation:
   - Clean the surface to remove dirt, grease, or rust.
   - Sand or wire brush to remove rust or flaking paint.
   - Use a metal degreaser if necessary.
2. Priming (often necessary):
   - Apply a metal primer to ensure paint adhesion and prevent rust.
3. Painting:
   - Apply paint suitable for metal, using a brush or spray.
   - Follow the manufacturer's instructions for application and drying times.

Part C

21. **Clear finishes** and **pigmented coatings** differ mainly in how they affect the wood’s look and the level of protection they provide.

Clear finishes keep the **natural grain and color** of the wood visible. They add a transparent protective layer that prevents moisture damage and surface wear. Common examples include **varnish, lacquer, shellac, and polyurethane**. For instance, **varnish** is often used on **wooden furniture and flooring** to enhance the grain

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and give a rich, glossy finish. **Lacquer** is popular for **decorative furniture or interior panels** because it dries quickly and provides a smooth surface. These finishes are chosen when the goal is to **highlight the wood's beauty** rather than cover it.

Pigmented coatings, in contrast, contain color pigments that **hide the natural wood grain** and give an even-colored surface. They are better for protecting wood from **sunlight, rain, and wear**, especially in outdoor settings. Examples include **paint, enamel, and opaque wood stains**. For example, **enamel paint** is often used on **doors and window frames** because it forms a hard, glossy, and weather-resistant surface. **Opaque stains** are commonly used on **garden furniture or wooden fences** to provide both color and protection from the elements.

In short, clear finishes enhance and show off the wood's natural character, while pigmented coatings focus on **color uniformity, durability, and protection**, especially for outdoor or high-use surfaces.

22. **Oil Finishes**

Oils penetrate wood fibers, bringing out natural grain patterns and providing a warm, natural look. Common types include **linseed oil**, **tung oil**, and **teak oil**. They offer moderate water resistance and are easy to apply with a cloth or brush. Multiple coats enhance depth and can be buffed for a subtle sheen. Oil finishes generally do not form a hard protective surface, so they may require reapplication over time.

Wax Finishes

Wax provides a soft, lustrous surface and is often used over other finishes for added shine. It protects against moisture and minor scratches but is not highly durable. Beeswax and carnauba wax are popular. Application involves rubbing the wax onto the surface and buffing to achieve a sheen. Wax finishes are suitable for furniture and decorative items rather than high-traffic surfaces.

Varnishes

Varnish forms a hard protective coating and is available in types such as **polyurethane**, **spar varnish**, and **aliphatic varnish**. Varnishes provide excellent resistance to water, heat, and wear. They are applied in thin, even coats with a brush, and sanding between coats ensures smoothness. Varnishes can be glossy, semi-gloss, or satin. Polyurethane is widely favored for its durability, especially on floors and tabletops.

Lacquers

Lacquer dries quickly and produces a durable, smooth finish with varying levels of gloss. It is traditionally dissolved in solvents and sprayed or brushed onto wood. Lacquers are popular in furniture manufacturing due to their high clarity and fast curing times. However, they require proper ventilation during application because of strong fumes.

Shellac

Shellac is a natural resin obtained from lac insects, dissolved in alcohol. It provides a warm tone and enhances the wood's natural beauty. Shellac dries quickly and polishes to a high gloss, but it is less resistant to water and heat. It is commonly used on fine furniture, antiques, and musical instruments.

Wood Stains

Stains are pigments or dyes that penetrate the wood surface, altering its color without forming a protective layer. They enhance the grain while providing coloration. Stains are often followed by a protective finish like varnish or polyurethane for durability. Oil-based and water-based stains are common, each with unique penetration and drying characteristics

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